Indian Airline Revenue and Profit Analysis

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March 2026

Introduction

- India's aviation industry is one of the **fastest-growing markets** in the world.
- Millions of passengers fly every year for business, tourism and personal travel.
- Airlines keep expanding routes and fleet capacity to meet demand.
- But operations cost a lot — fuel, maintenance, staff and airport charges.
- So **revenue growth does not always mean profit growth**.

Strong revenue, yet unstable profits — that is what we studied.

Airline Industry Overview

Revenue Sources

- Passenger tickets
- Cargo services
- Ancillary fees (baggage, onboard)

Cost Components

- Fuel (ATF) — largest cost
- Aircraft maintenance
- Airport charges & crew salaries

What Makes This Industry Unique

- **Capital-intensive** — heavy fixed costs from aircraft leases and depreciation.
- **Perishable inventory** — empty seat at takeoff = zero revenue, full cost.
- **Demand volatility** — seasons, economy and events affect occupancy regularly.

Problem Statement

- Passenger numbers in India have grown steadily.
- Airlines earn high revenue — but profits remain unstable.
- High costs eat into earnings, even in good years.

Research Question

Why do airlines earn high revenue but still face losses?

Key Numbers

IndiGo Revenue	**Rs. 3.08M**
IndiGo Profit	+Rs. 30K only
Air India Rev	**Rs. 2.12M**
Air India Loss	–Rs. 525K
Industry Total	**Rs. 6.31M**
Net Loss	–Rs. 184K

Objectives

01 Analyze airline **revenue performance** using DGCA data.

02 Study the link between **passenger demand** and financial results.

03 Evaluate **operational efficiency** using Load Factor, Passenger Number , Aircraft Number.

04 Identify key **factors affecting profitability**.

05 Generate **insights** to support revenue optimization.

Proposed Solution — Analytical Framework

Linking **traffic data** with **financial data** through four steps:

1. Passenger Demand	2. Capacity Utilisation	3. Operating Costs	4. Overall Profitability
How many passengers? Are seats filling up?	Is the right capacity deployed at the right time?	What does it cost to operate? Where are the gaps?	Is revenue turning into stable profit?

Demand $\rightarrow$ *Capacity* $\rightarrow$ *Cost* $\rightarrow$ *Profitability*

Methodology

01 Data Collection
DGCA reports on passengers, aircraft operations, revenue, expenses and profit.

02 Data Cleaning
Removed duplicates, fixed missing values, standardized formats across all airlines and years.

03 Exploratory Data Analysis
Found trends in revenue, costs, load factor and profits from FY 2019–2025.

04 KPI Calculation
Computed ASK, RPK, Load Factor, Revenue per Passenger and Cost per Passenger.

05 Dashboard Development
Built Power BI dashboards: Industry overview, IndiGo and Air India deep-dives.

Dataset Design — DGCA Data (FY 2019–2025)

Source: DGCA, India **Airlines:** IndiGo, Air India, SpiceJet, Air India Express, Akasa Air, Star Air, Alliance Air

Operational Dataset

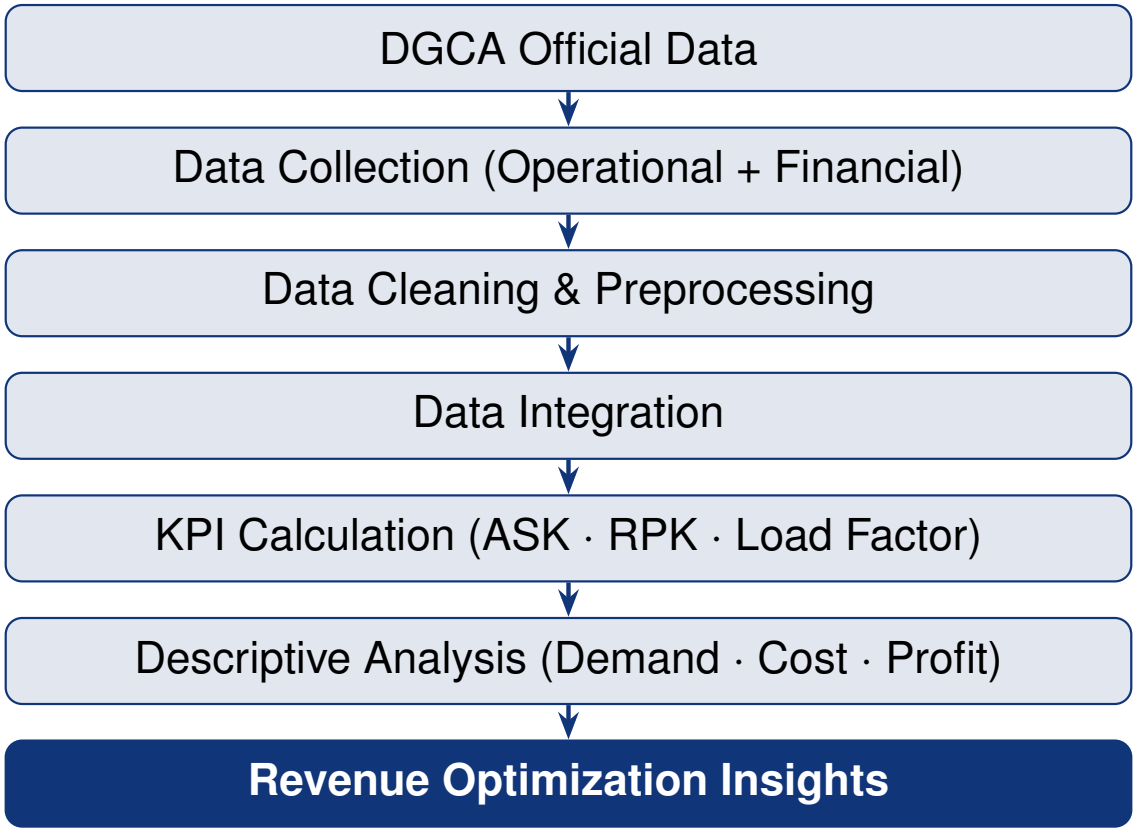
- Passenger Numbers
- Aircraft Operations
- ASK — Available Seat Kilometres
- RPK — Revenue Passenger Kilometres
- Load Factor (%)

Financial Dataset

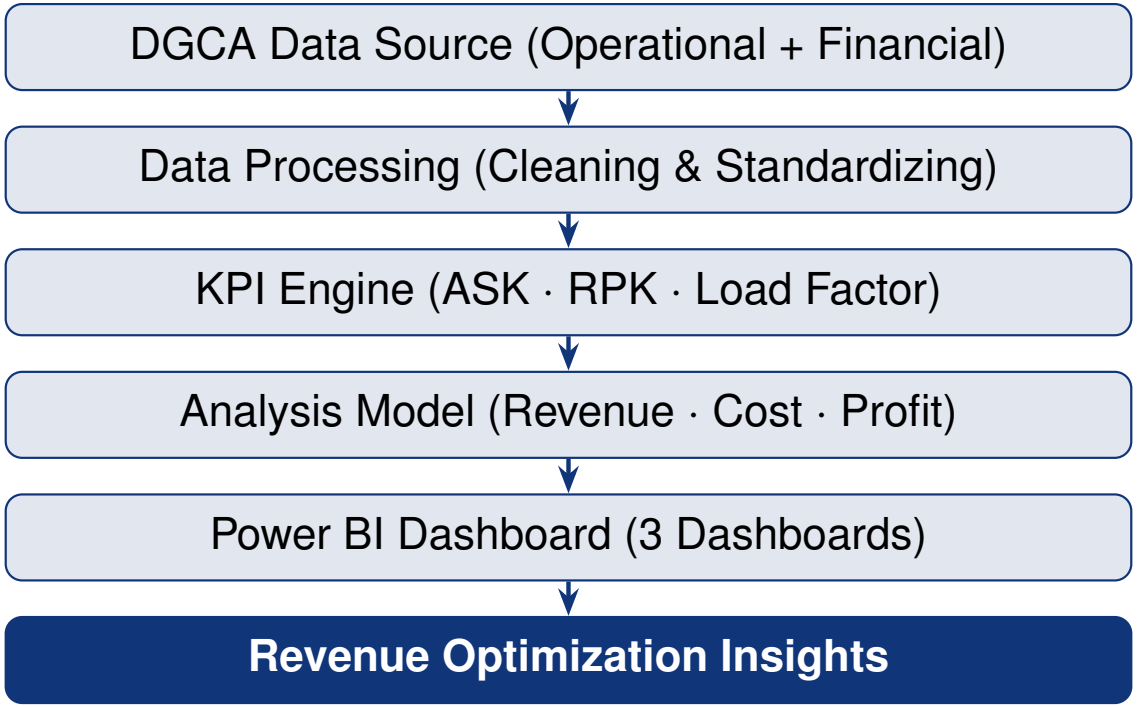
- Total Operating Revenue
- Total Operating Expenses
- Net Profit / Loss
- Expense Breakdown (8 categories)
- Revenue & Cost per Passenger

Airline name is the key linking both datasets.

Analytical Workflow



System Block Diagram



EDA Observations

- **Profit Fluctuated** — Sharp losses during the disruption period.
- **Revenue Recovered** — Passenger demand bounced back strongly.
- **Expenses Also Rose** — Costs increased along with revenue.
- **Profit Lagged** — Revenue recovery did not lead to profit recovery.

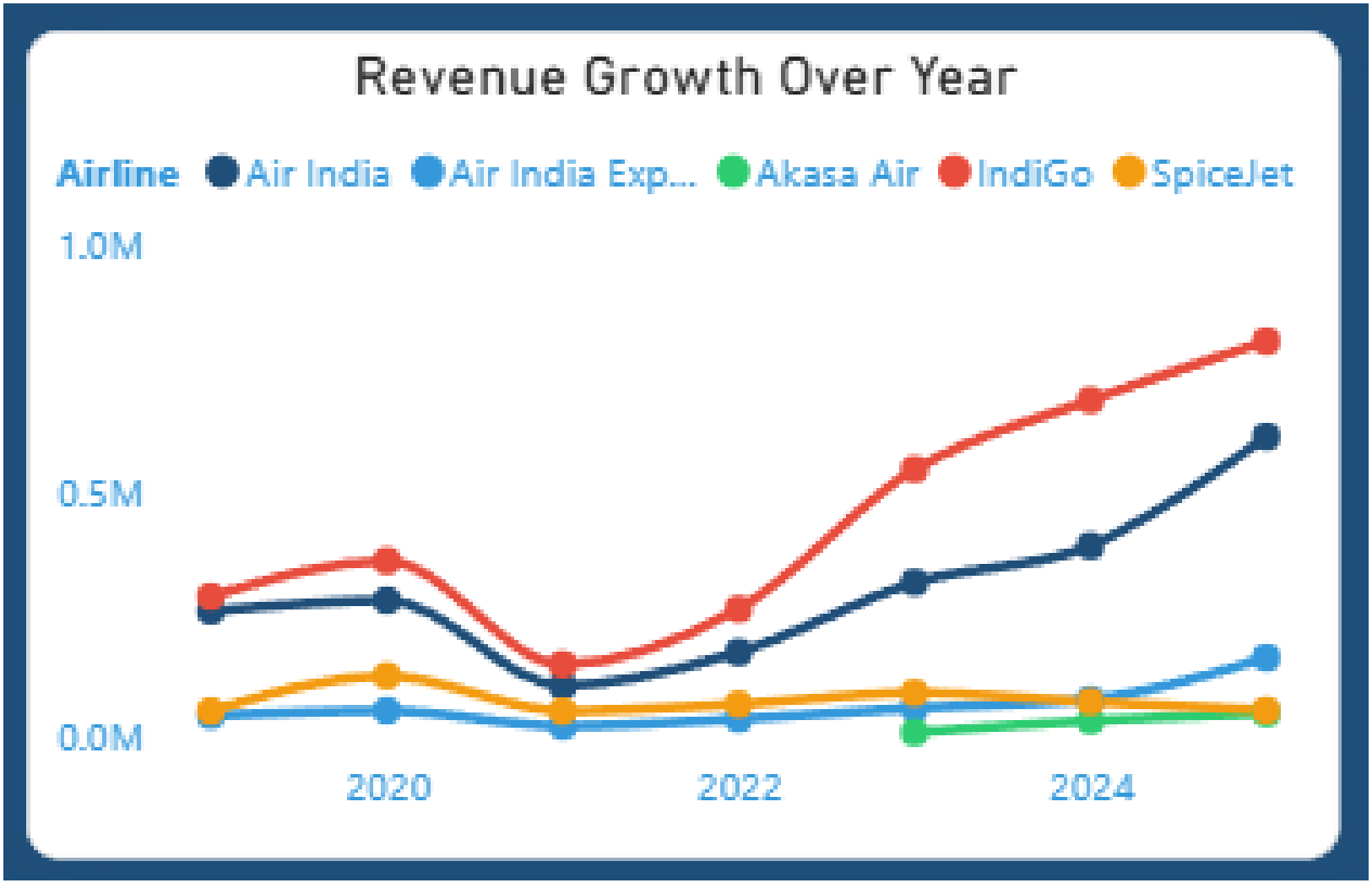
Key Insight

Revenue grew — but **costs grew at the same pace**.

More passengers alone does **NOT** guarantee profit.

This pushed us deeper into cost analysis.

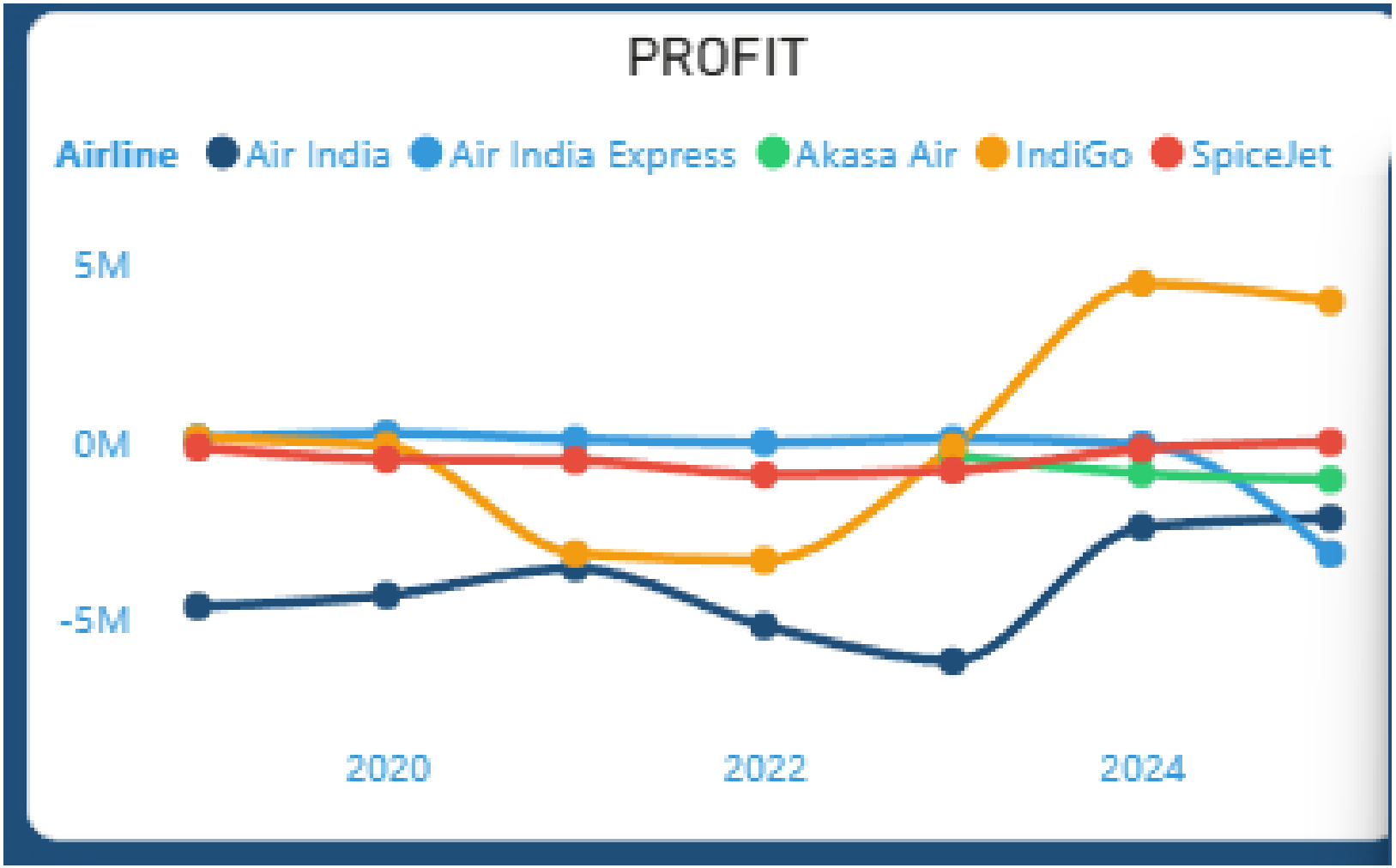
Revenue Growth Trend (FY2019–FY2025)



Insight:

- Airline revenue increased steadily with rising passenger demand.
- Recovery accelerated after disruption years.

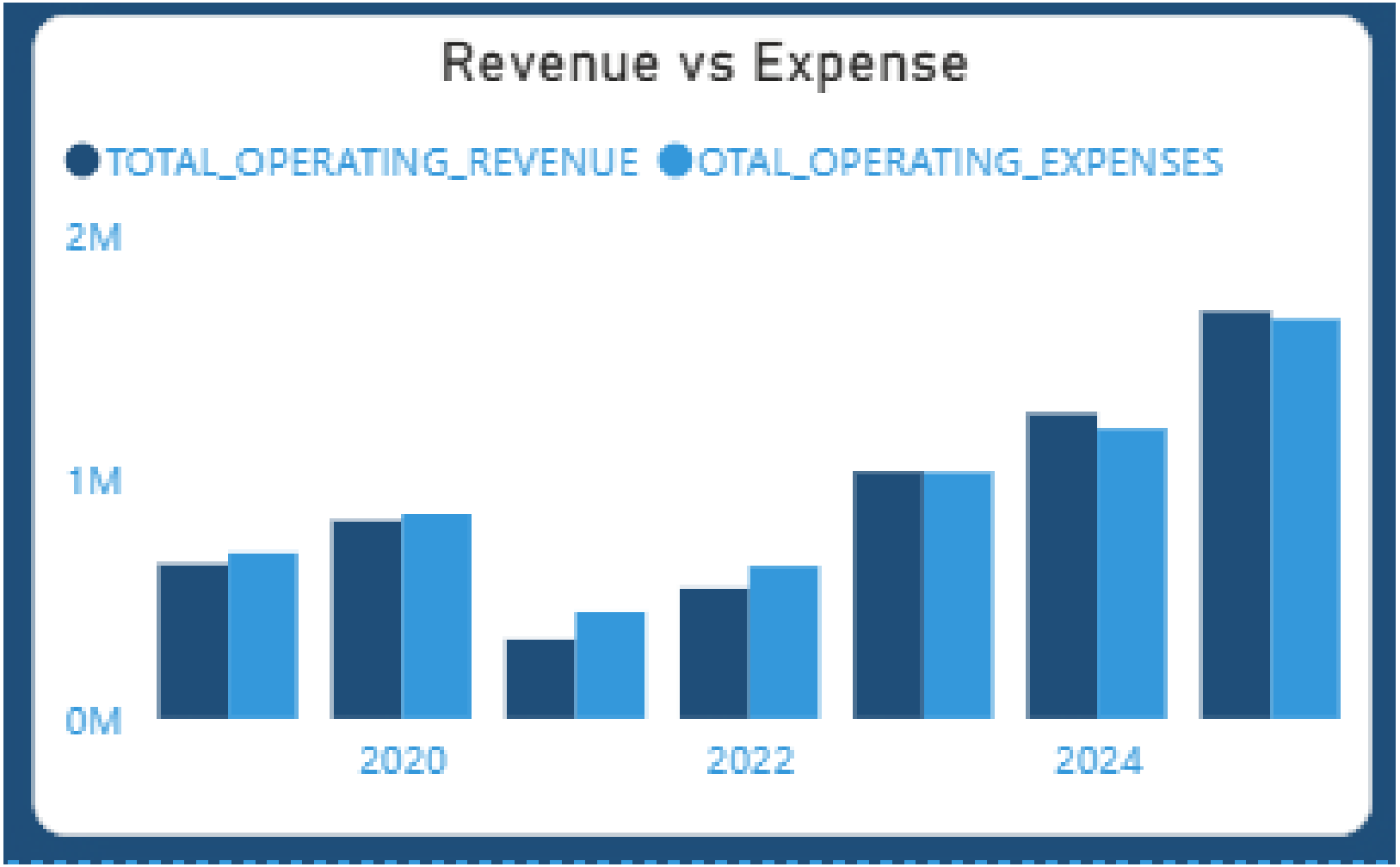
Profit Growth Trend



Insight:

- Profit did not grow consistently with revenue.
- Airlines experienced significant fluctuations in profitability.
- High operating costs limited financial recovery.

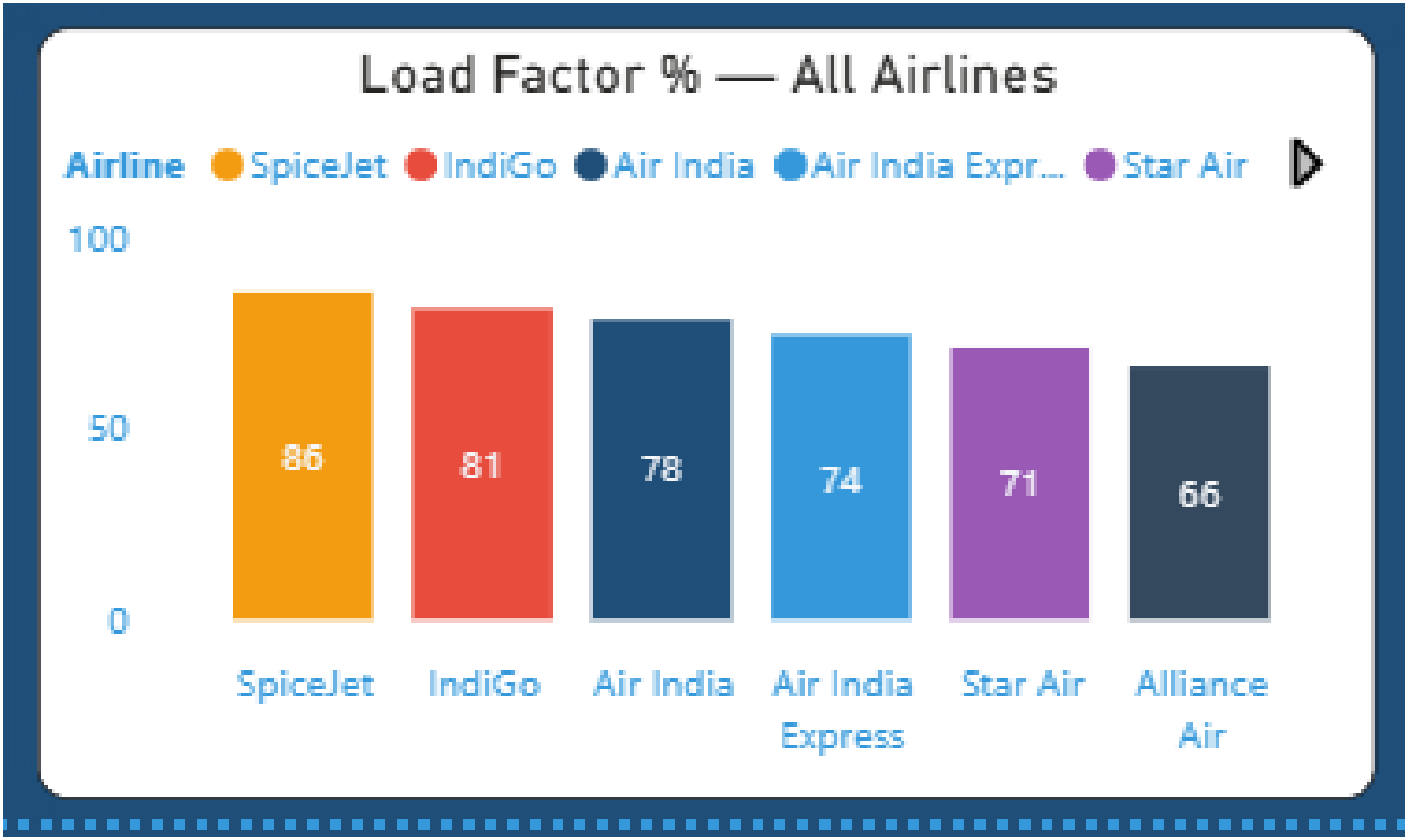
Revenue vs Profit Comparison



Insight:

- Operating expenses absorbed most of the revenue gains.
- This explains why airlines generate large revenue but still face losses.

Seat Load Factor Trend



Insight:

- Load factor measures seat utilisation efficiency.
- Higher load factors improve revenue per flight.
- Airlines with better seat utilisation tend to achieve stronger financial performance.

IndiGo vs Air India Comparison

Metric	Industry Total	IndiGo	Air India
Total Aircraft Operations	6 Million	4 Million	977 Thousand
Total Passengers Carried	828 Million	531 Million	138 Million
Total Revenue	■6.31 Million	■3.08 Million	■2.12 Million
Total Expenses	■6.50 Million	■2.92 Million	■2.35 Million
Profit / Loss	-■184,000	+■30,100	-■525,000
Load Factor	—	81%	78%
Profit Margin (FY25)	~-3%	~+8%	~-15%
Rev/Pax > Cost/Pax?	Industry: NO	YES (FY23+)	NO
Flight Ops % of Expenses	46.88%	~47%	~44%
Data Period	FY 2019–2025	FY 2019–2025	FY 2019–2025

IndiGo vs Air India — Key Insights

- **Scale Advantage** IndiGo operates nearly **4 million flights** and carried **531 million passengers**, far higher than Air India.
- **Revenue Similarity, Profit Difference** Both airlines generate large revenue, but IndiGo remains profitable while Air India reports heavy losses.
- **Operational Efficiency** IndiGo maintains a slightly higher **load factor (81%)** compared to Air India (**78%**), improving seat utilisation.
- **Cost Control** IndiGo keeps **revenue per passenger higher than cost per passenger**, while Air India struggles with higher operating costs.
- **Profitability Gap** IndiGo achieved about **+8% profit margin**, whereas Air India recorded nearly **-15% margin**.

Operational efficiency and cost control explain the profitability gap between the two airlines.

Key Findings

01 Passenger Growth $\neq$ Profit Growth
Demand recovered but profit lagged behind.

02 Rising Costs Kill Margins
Fuel costs rose with revenue, leaving almost no profit gap.

03 Cost/Pax $\approx$ Rev/Pax = Danger Zone
When both are nearly equal, any shock pushes the airline into a loss.

04 Efficiency Drives Recovery
IndiGo (better cost control) recovered faster than Air India.

05 Seasonality Worsens the Problem
Too many flights in low-demand seasons increases costs faster than revenue.

Project Outcomes

Integrated Dataset

Clean, unified DGCA data covering 7 airlines across FY 2019–2025.

3 Power BI Dashboards

Industry overview + IndiGo and Air India deep-dives with year and airline filters.

KPI Framework

Computes Load Factor, ASK, RPK, Revenue and Cost per Passenger.

Optimization Insights

5 key profitability factors identified with actionable recommendations.

Future Scope

Predictive ML Models

Forecast passenger demand by route and season for better capacity planning.

Dynamic Pricing Simulation

Model ticket pricing strategies to maximize revenue per available seat.

Fuel Price Integration

Add ATF fuel price data to model cost changes and predict profit more accurately.

International Markets

Extend the framework to global airlines for benchmarking and comparison.

Conclusion

"Profitable airlines balance three things: Demand Forecasting, Capacity Management and Cost Efficiency."

Demand Forecasting	Capacity Management	Cost Efficiency
Predict passenger volumes by season and route. Price smartly to fill seats.	Fly the right number of flights at the right time. Avoid excess capacity.	Keep cost per passenger below revenue per passenger at all times.

IndiGo = Proof | **Air India** = The Warning

Thank You Questions?